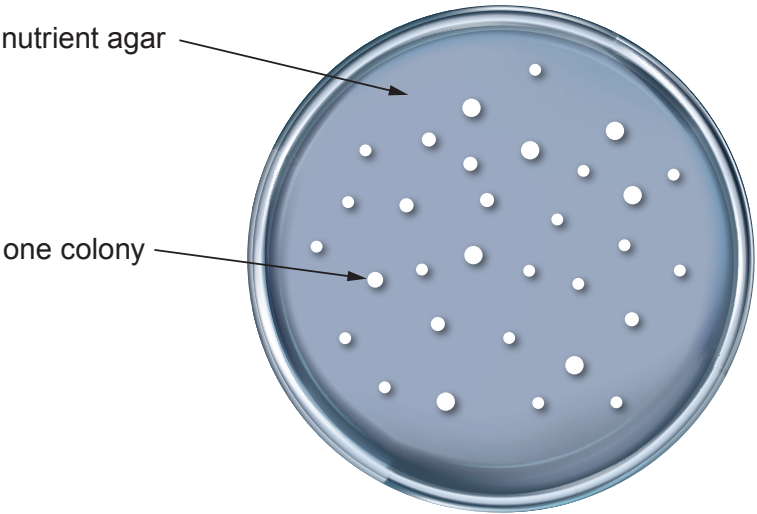


6. Environmental health officers estimated the number of bacteria in a milk sample from a cafe. The limit for the total number of bacteria in a sample of milk which is considered to be safe for human consumption is 1.0×10^5 bacteria per cm^3 .
- Using aseptic techniques, they diluted the sample by a factor of 10 000.
 - They plated 1 cm^3 of the diluted sample onto nutrient agar.
 - The lid of the plate was secured with tape.
 - The plate was incubated at 37°C for 2 days.
 - The result is shown in **Image 6.1**.

Image 6.1



- (a) (i) State the assumption that must be made when calculating the number of bacterial cells present in the original sample. [1]

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- (ii) I. Calculate the number of bacteria in 1 cm^3 of the original sample taken by the environmental health officers. **Write your answer in standard form.** [3]

Number of bacteria =

- II. Conclude whether the milk sample was safe for human consumption. Explain your answer. [1]

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(b) Suggest why the plates were incubated at 37 °C. [1]

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(c) (i) State why the lids of the agar plates were secured with tape. [1]

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(ii) State **two** other precautions that should have been taken to ensure aseptic technique. [2]

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